

From History to Predictions: Machine Learning Approaches to Epilepsy Diagnosis

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Epilepsy is a chronic neurological disorder caused by recurring seizures due to abnormal activity amongst nerve cells. Due to the vastness of types of epilepsy and symptoms, there are many different medications that are meant to manage the symptoms and prevent seizures. The aim of this project is to use clinical and seizure-related features to build, evaluate, and interpret machine learning models that predict epilepsy types and identify the symptoms that most strongly influence classification.

The dataset utilized for this project was the Epilepsy Disorder Dataset from Kaggle. The dataset has 30 columns and 4,856 observations. The target column in epilepsy type with 6 possible outcomes: Absence, Normal, Complicated, Focal, Generalized and Unknown. Of the 29 predictors, 20 are binary categorical variables, 4 are multi-level categorical, and 5 are numeric, looking into the patient's physiology like height and weight and length of their seizures. Since the multi-categorical variables were stored as strings, they were one-hot encoded to prepare them for modeling and prediction. The numeric variables were standardized to ensure fair weighting for models sensitive to Euclidean distance, while binary categorical variables were encoded with 1/0.

Data visualization was performed to examine distributional disparities and uncover underlying trends in the dataset. The response variable was unevenly distributed across the six epilepsy types, with generalized epilepsy representing the largest proportion of cases. In addition, males showed a higher rate of epilepsy compared to individuals who do not identify as male. Age-based visualizations suggested that the dataset may have been collected through stratified random sampling, as each age group contained a similar number of observations. The distribution also showed that individuals with absence epilepsy tended to be younger, which aligns with known patterns in clinical and epidemiological research.

A UMAP and correlation plot were also plotted to identify clusters or relationships between the variables. The UMAP identified 3 big clusters, mainly that of Absence, Focal and Generalized epilepsy. The other three categories were scattered throughout and did not illustrate any distinct clusters. The

correlation plot illustrated slight correlation with the target response and more correlation between variables, indicating that a blend of response variables is likely needed to predict the response.

Next, multiple models were fitted to classify the data. A total of seven algorithms, Gradient Boosting, CatBoost, XGBoost, Logistic Regression, Random Forest, SVM, and k-NN, were trained and compared. Model performance was evaluated using weighted F1 and overall accuracy. The weighted F1 metric was selected due to the uneven distribution of the response classes identified earlier in the project. Among all models, Gradient Boosting performed the best, achieving a weighted F1 of 0.823 and an accuracy of 0.8143. Its shallow tree structure handled the moderate-sized dataset effectively, whereas more complex tree-based models such as XGBoost and CatBoost tended to overfit. In contrast, k-NN showed the weakest performance, with a weighted F1 of 0.7233 and an accuracy of 0.6810, likely due to the large number of binary predictors; Euclidean-distance-based methods struggle when most features are sparse or high-dimensional.

Given that gradient boosting performed the best, its model parameters were then tuned. XGBoost was also tuned to determine if tuning could possibly produce a better model than gradient boosting. After tuning, gradient boosting still outperformed XGBoost with a weighted F1 of 0.814 and accuracy of 0.824. The best model parameters obtained were: $n = 200$, learning rate = 0.05, depth = 3.

Feature selection was then performed on the gradient boosting model to identify the predictors most influential in classifying epilepsy type. The top 15 features accounted for approximately 98.3% of the model's predictive signal. Notably, four of these features were one-hot encoded levels of the same variable, seizure type, highlighting its critical role in determining both the type and presence of epilepsy. The reduced model achieved a weighted F1 of 0.793 and an accuracy of 0.807. This modest ~0.02 decrease from the full model suggests that a relatively small subset of features captures the majority of predictive power. As patient sample sizes grow, this reduced model could be used to efficiently and reliably predict epilepsy type while offering faster computation and simpler clinical interpretation.

Overall, the project illustrated that a compact set of clinically meaningful symptoms is sufficient for accurate epilepsy classification using Gradient Boosting, and can be implemented in the future to support rapid, reliable diagnostic decision-making.